

Practice Quiz: Cognition (Neuroscientific Frontiers)

Name: __ Date: __

Part A: Multiple Choice

1. The vmPFC's role in Active Inference decision-making is: A) Motor execution B) Computing pragmatic value — estimating expected reward for each candidate policy C) Sensory processing D) Language production
2. The ACC signals decision difficulty, which corresponds to: A) High reward B) Low precision of the posterior over policies — multiple policies have similar expected free energy C) Fast reaction time D) Automatic processing
3. Tonic dopamine levels control: A) Vision B) Policy precision (γ) — the confidence of policy selection, mapping onto decisiveness vs. randomness C) Memory storage D) Heartbeat
4. Working memory capacity (~4 items) is constrained by: A) Brain size B) The number of gamma bursts that fit within one theta cycle — a biophysical limit on simultaneous belief maintenance C) Available oxygen D) Education level
5. Hippocampal replay implements: A) Dreaming B) Mental simulation — evaluating candidate policies by replaying past trajectories at accelerated speed C) Forgetting D) Reflexes
6. The exploration-exploitation trade-off is resolved by EFE because: A) EFE only values reward B) EFE automatically balances pragmatic value (exploitation) with epistemic value (exploration) in a single objective C) EFE ignores information D) Exploration is always better
7. Frontopolar cortex (BA10) implements: A) Basic reflexes B) Counterfactual reasoning — evaluating policies not currently being executed and maintaining alternative strategies C) Pain processing D) Balance

Part B: Essay Questions

1. Compare Active Inference's account of decision-making with classical reinforcement learning (RL). Active Inference uses expected free energy; RL uses expected cumulative reward. What are the key differences? What phenomena can Active Inference explain that RL cannot (and vice versa)? Reference specific experimental paradigms. (400 words)
2. Design a neuroimaging experiment to dissociate pragmatic and epistemic value in the brain. Specify: (a) a decision paradigm where pragmatic and epistemic value can be independently varied, (b) the predicted neural activations for each, (c) the critical contrast that would demonstrate dissociation. (400 words)
3. How does the dopamine-as-precision account explain comorbidity between psychiatric conditions and decision-making deficits? Choose two conditions (e.g., addiction and impulsivity; depression and indecisiveness; schizophrenia and jumping to conclusions) and provide a unified precision-based explanation. (400 words)